

NJIT – CS 675 Machine Learning – Fall 2026

Three pictures

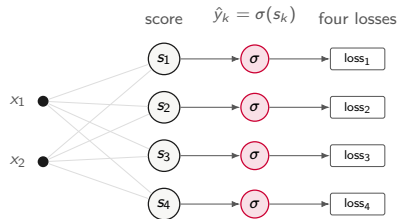
hyperparameters – one-vs-rest against softmax – nearest neighbours as attention

Hyperparameters

What is a hyperparameter?

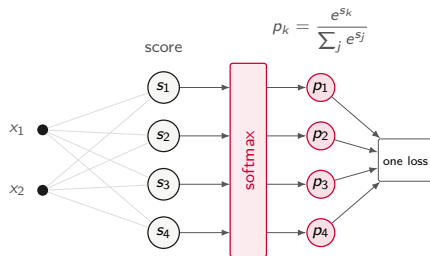
Four units side by side, or one softmax layer

One-vs-rest: k independent units



trained **one at a time**, each on "is it class k ?"
outputs need not add to 1; predict by argmax

Softmax: k units linked by one layer



trained **together** on the cross-entropy $-\ln p_{true}$
outputs add to 1; predict by argmax

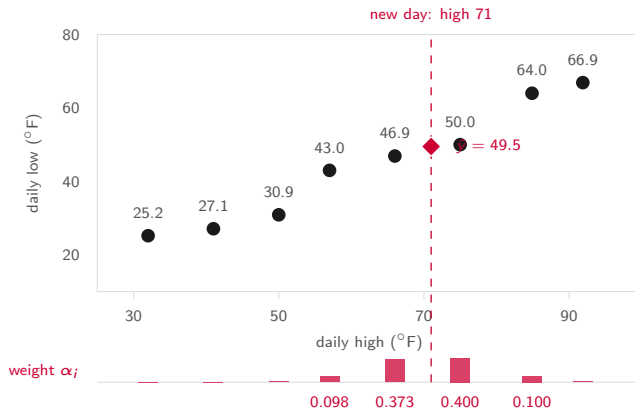
	separate units	softmax layer
output	$\sigma(s_k)$	p_k
computed from	its own score	all k scores
loss	one per unit	one for the model
trained	separately	jointly

Same inputs, same units, **different weights**: on the Newark seasons the two models name a different season on 160 of 1,457 validation days.

Notes: 5.5 (one-vs-rest), 16.2 and 16.3 (softmax, cross-entropy).

A day the two disagree on: <https://ikoutis.github.io/courses/cs675/f26/four-units.html>

Nearest neighbours, as attention



A regression by memory: eight remembered days (x_i, y_i) and a new day with high x .

score, from the distance $s_i = -\frac{d(x, x_i)^2}{2\tau^2}$

softmax: weights $\alpha_i = \frac{e^{s_i}}{\sum_j e^{s_j}}$

answer $\hat{y} = \sum_i \alpha_i y_i$

k -NN regression: $\alpha_i = 1/k$ for the k nearest days and 0 for the rest. Attention: every remembered day votes, with a weight that falls with its distance; τ plays the role of k .

Here $\tau = 8$. The three nearest days average 53.6; the least-squares line says 53.4. Notes: 2.4, 2.12, 16.2.